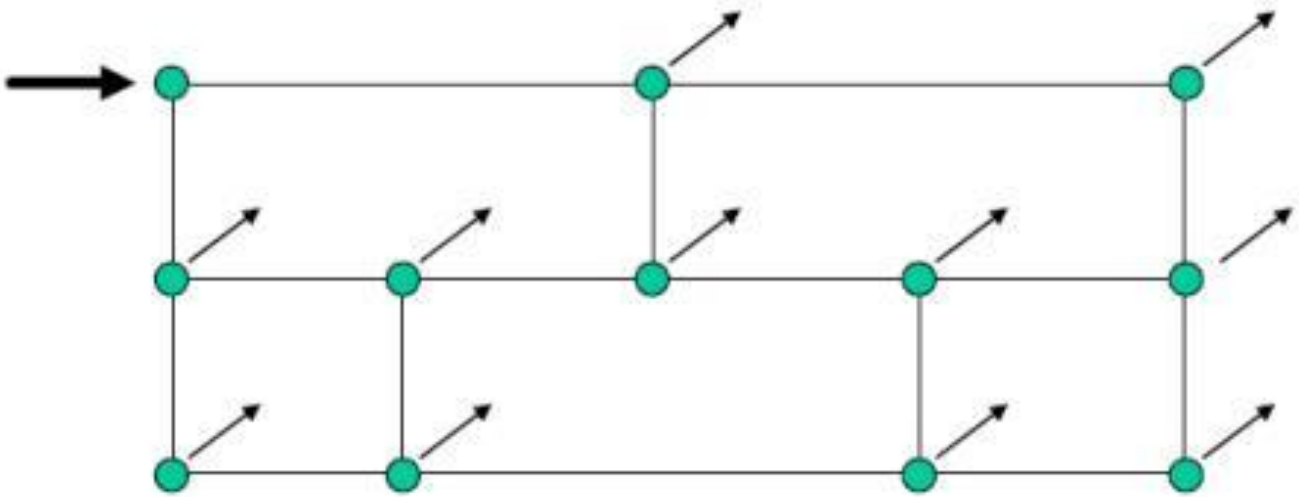
Pipe Network Analysis



Pipe Networks

- **A hydraulic model** is useful for examining the impact of design and operation decisions.
- **Simple systems,** such as those discussed in last chapters can be solved using a hand calculator.
- However, more **complex systems** require more effort even for *steady state conditions*, but, as in simple systems, the flow and pressure-head distribution through a water distribution system must *satisfy the laws of conservation of mass and energy*.

Pipe Networks

- **The equations to solve Pipe network must satisfy the following condition:**

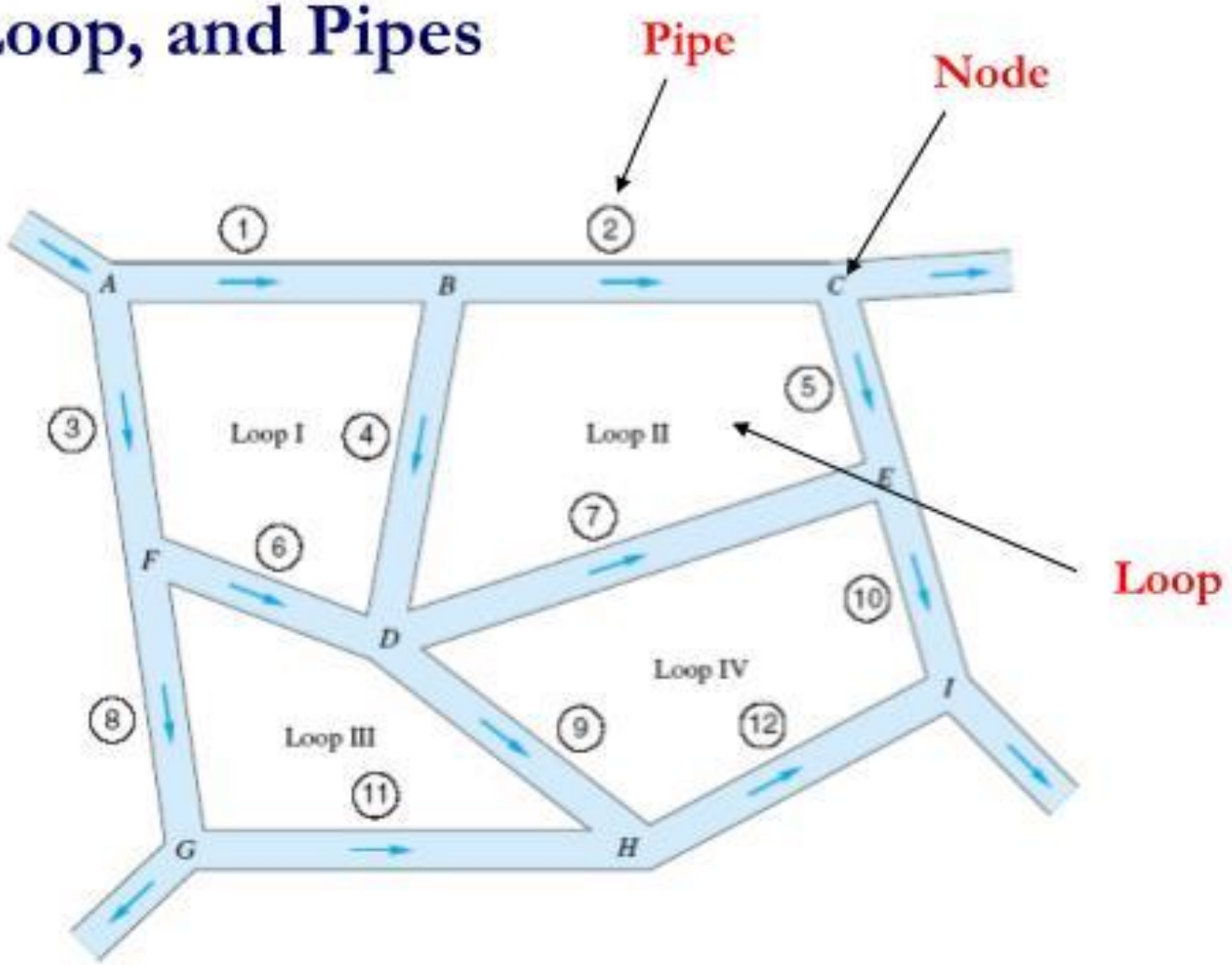
- The net flow into any junction must be zero

$$\sum Q = 0$$

- The net head loss a round any closed loop must be zero. The HGL at each junction must have one and only one elevation
- All head losses must satisfy the Moody and minor-loss friction correlation

Pipe Network Analysis

Node, Loop, and Pipes



Hydraulic Analysis

After completing all preliminary studies and layout drawing of the network, one of the methods of hydraulic analysis is used to

- Size the pipes and
- Assign the pressures and velocities required.

Hydraulic Analysis of Water Networks

- The solution to the problem is based on the same basic hydraulic principles that govern simple and compound pipes that were discussed previously.
- The following are the most common methods used to analyze the Grid-system networks:

1. Hardy Cross method.
2. Sections method.
3. Circle method.
4. Computer programs (Epanet,Loop, watercad..)

(1). The Hardy Cross Method

- The earliest systematic method of network analysis (Hardy-Cross Method) is known as the head balance or closed loop method.

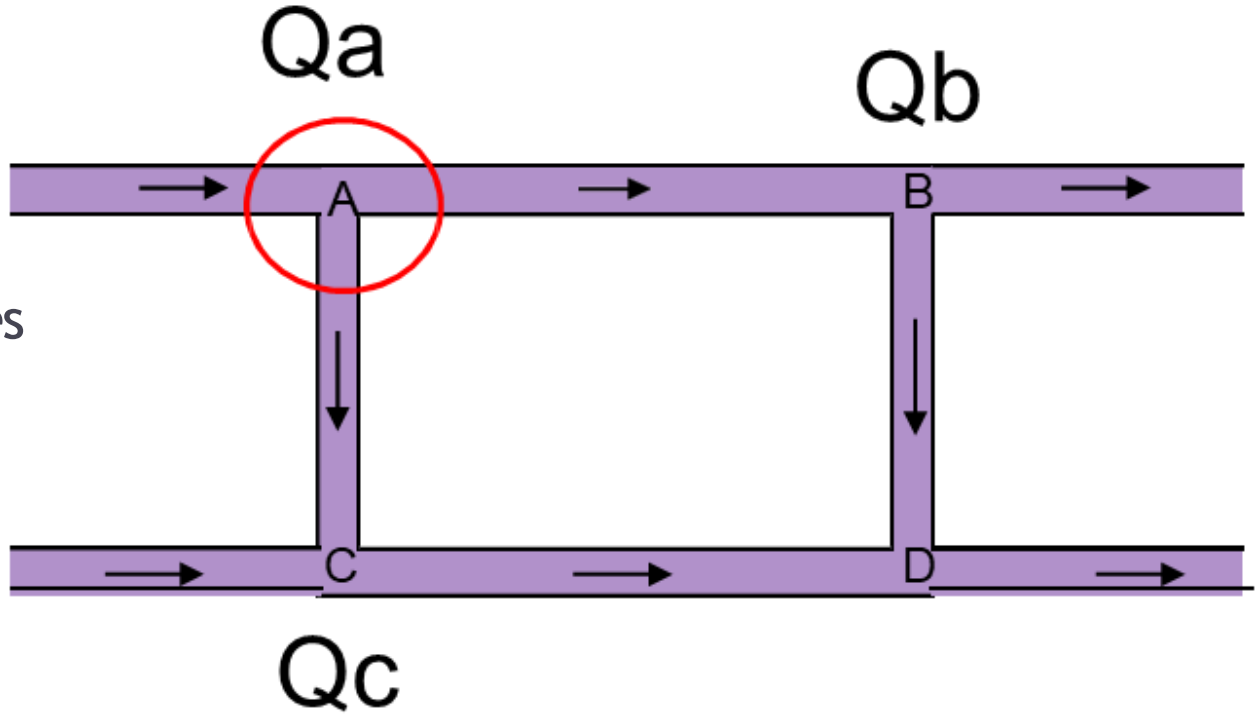
- This method is applicable to system in which pipes form closed loops. The outflows from the system are generally assumed to occur at the nodes junction.

- For a given pipe system with known outflows, the Hardy-Cross method is an iterative procedure based on initially iterated flows in the pipes.

- At each junction these flows must satisfy the continuity criterion, i.e. the algebraic sum of the flow rates in the pipe meeting at a junction, together with any external flows is zero.

(1).The Hardy Cross Method

- The method is based on



- **Continuity Equation:**
  - Inflow = outflow at nodes

$$Q_a = Q_b + Q_c$$

- **Energy Equation:**
  - Summation of head loss in closed loop is zero

$$\sum h_l(loop) = 0 \Rightarrow \sum K(Q + \Delta Q)^n = 0$$

(1).The Hardy Cross Method

- The relationship between head loss and discharge must be maintained for each pipe

  - Darcy-Weisbach Equation

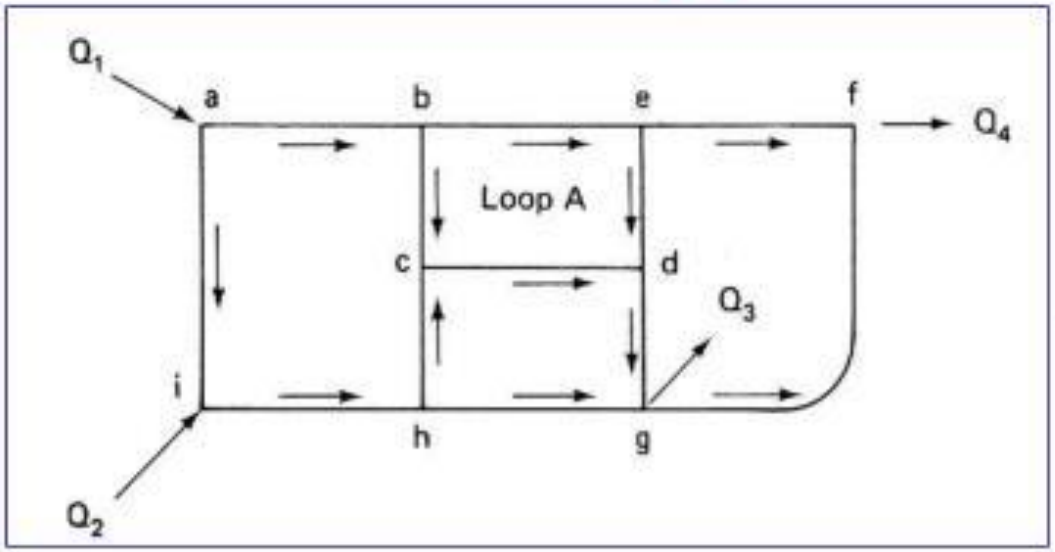
$$h_l(\,pipe) = KQ^n \quad n = 2 \quad K = \frac{8\,fL}{g\pi^2 D^5}$$

  - Exponential friction formula Hazen-Williams

$$h_l(\,pipe) = KQ^n \quad n = 1.85 \quad K = \frac{10.67}{C^{1.85} d^{4.87}}$$

Pipe Network Analysis

Hardy Cross Method



This method based on:

$$\sum_{Junction} Q = 0 \qquad \sum_{Loop} h_f = 0$$

1- A distribution of flows in each pipe is estimated such that the total inflow must be equal to the outflow at each junction throughout the network system

The interflow in the network has $+ve$ sign

The outflow from the network has $-ve$ sign

Pipe Network Analysis

2- Neglect Minor loss

3- In each loop $\sum_{\text{Loop}} h_f = 0$

4- If the direction of flow is clockwise it take +ve sign, otherwise it take –ve sign

5- If $\sum_{\text{Loop}} h_f = 0$ the flow is correct other wise, the assumed flow must be corrected as the flowing:

Pipe Network Analysis

$$h_F = kQ^n \longrightarrow (1)$$

$n = 2 \Rightarrow Darcy, Manning$

$n = 1.85 \Rightarrow Hazen\ \ William$

$$Q = Q_o + \Delta \longrightarrow (2)$$

from 1 & 2

$$h_f = kQ^n = k(Q_o + \Delta)^n = k\left[Q_o^n + nQ_o^{n-1}\Delta + \frac{n(n-1)}{2}Q_o^{n-2}\Delta^2 + ...\right]$$

Neglect terms contains Δ^2 $\qquad h_f = kQ^n = k\left(Q_o^n + nQ_o^{n-1}\Delta\right)$

For each loop

$$\sum_{loop} h_F = \sum_{loop} kQ^n = 0$$

$$\therefore \sum kQ^n = \sum kQ_o^n + \sum nkQ^{(n-1)}\Delta = 0$$

Pipe Network Analysis

$$\Delta = \frac{-\sum kQ_o^n}{\sum nkQ_o^{(n-1)}} = \frac{-\sum h_F}{n\sum \frac{h_F}{Q_o}}$$

6- After calculation Δ correct Q_o and check $\sum_{\text{Loop}} h_f \approx 0$

(1). Hardy Cross Method

- **Problem Description**

- Network of pipes forming one or more closed loops

- Given
  - Demands @ network nodes (junctions)
  - d, L, pipe material,Temp, and P @ one node

- Find
  - Discharge & flow direction for all pipes in network
  - Pressure @ all nodes &HGL

(1). Hardy-Cross Method (Procedure)

1. Divide network into number of closed loops.

2. For each loop:

a) Assume discharge Q_a and direction for each pipe. Apply Continuity at each node, Total inflow = Total Outflow. Clockwise positive.

b) Calculate equivalent resistance K for each pipe given L, d, pipe material and water temperature.

c) Calculate $h_f = K\ Q_a^{\ n}$ for each pipe. Retain sign from step (a) and compute sum for loop $\Sigma\ h_f$.

d) Calculate $|h_f / Q_a|$ for each pipe and sum for loop $\Sigma|h_f / Q_a|$.

e) Calculate correction

$$\Delta Q = -\Sigma h_f / (n\Sigma|h_f/Q_a|).$$

NOTE: For common members between 2 loops both corrections have to be made. As loop 1 member, $\Delta Q = \Delta Q_1 - \Delta Q_2$. As loop 2 member, $\Delta Q = \Delta Q_2 - \Delta Q_1$.

f) Apply correction to Q_a, $Q_{new} = Q_a + \Delta Q$.

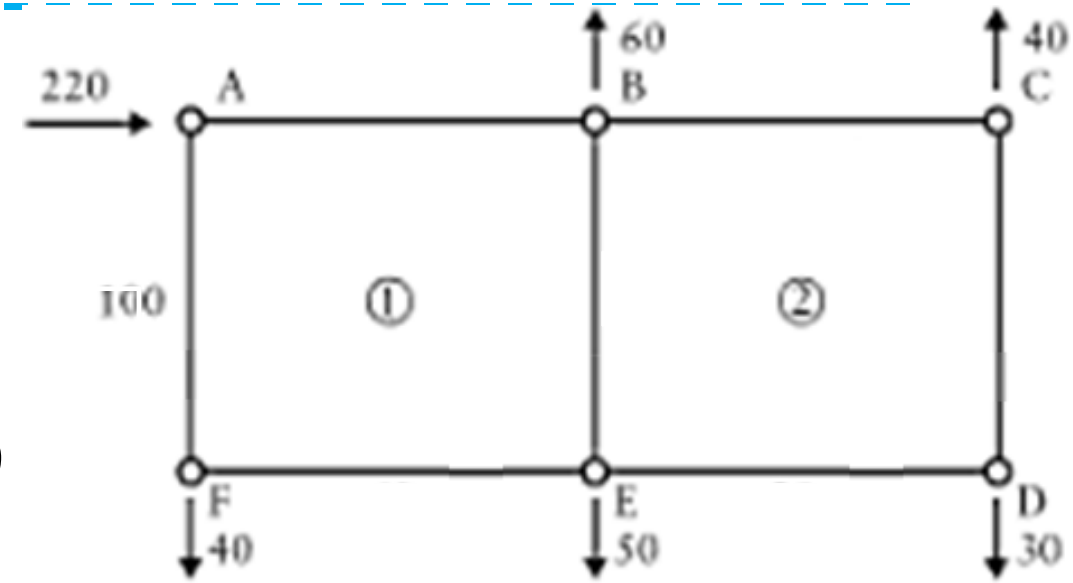
g) Repeat steps (c) to (f) until ΔQ becomes very small and $\Sigma h_f = 0$ in step (c).

h) Solve for pressure at each node using energy conservation.

Example: 1

Neglecting minor losses in the pipe, determine the flows in the pipes and the pressure heads at the nodes

(kinematic viscosity= $1.13 \times 10^{-6} m^2/s$)



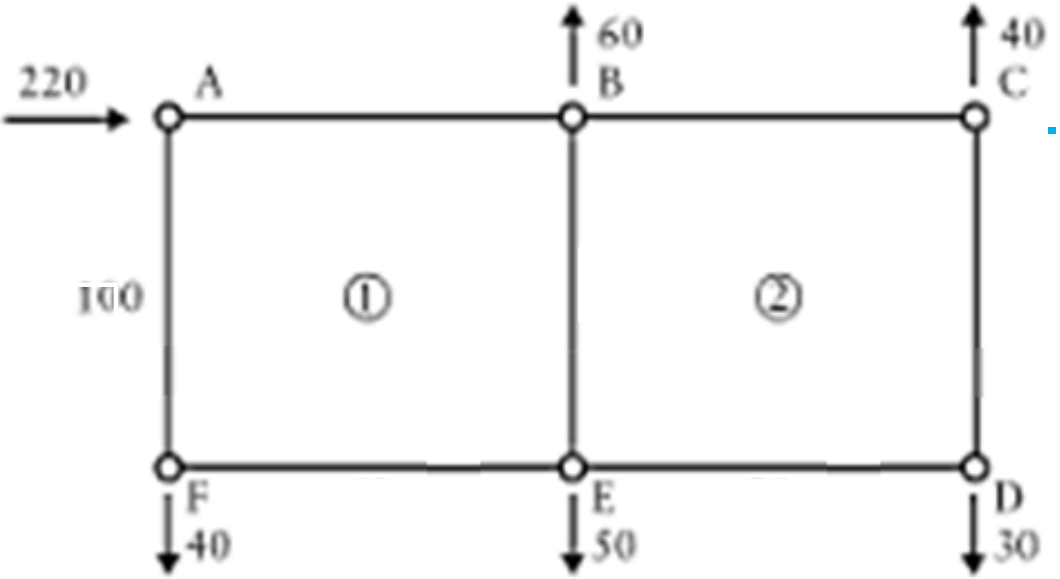
Data

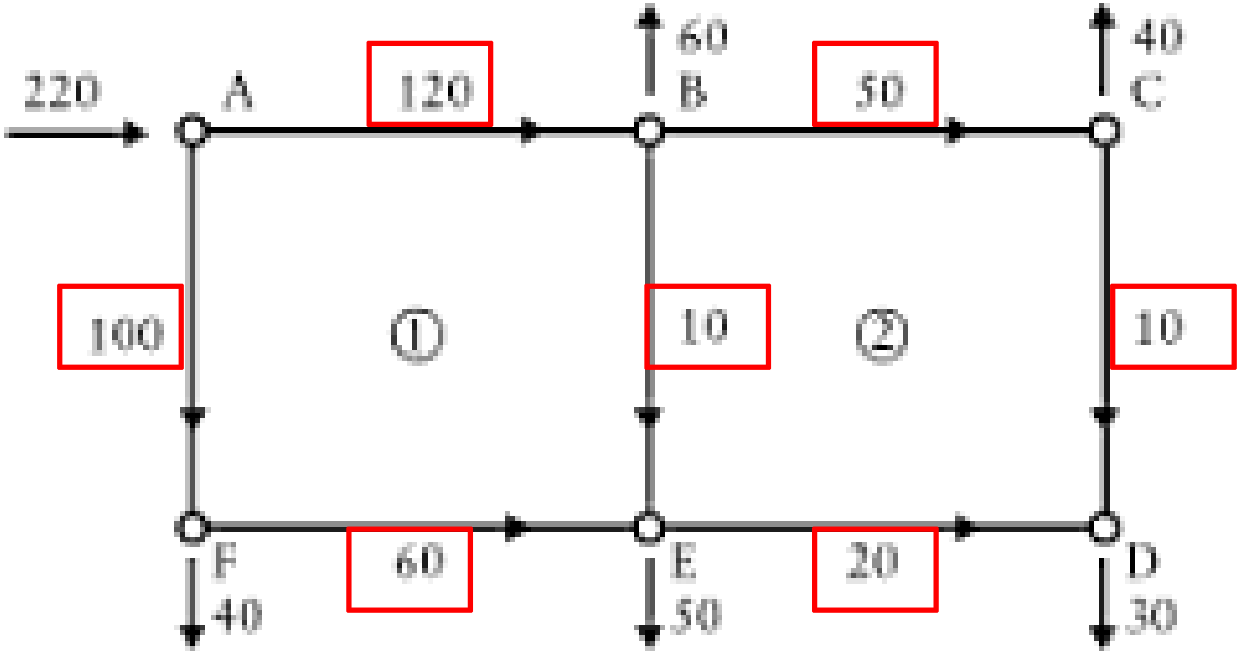
Pipe	AB	BC	CD	DE	EF	AF	BE
Length (m)	600	600	200	600	600	200	200
Diameter (mm)	250	150	100	150	150	200	100

Roughness size of all pipes = 0.06 mm
Pressure head elevation at A = 70 m o.d.

Elevation of pipe nodes

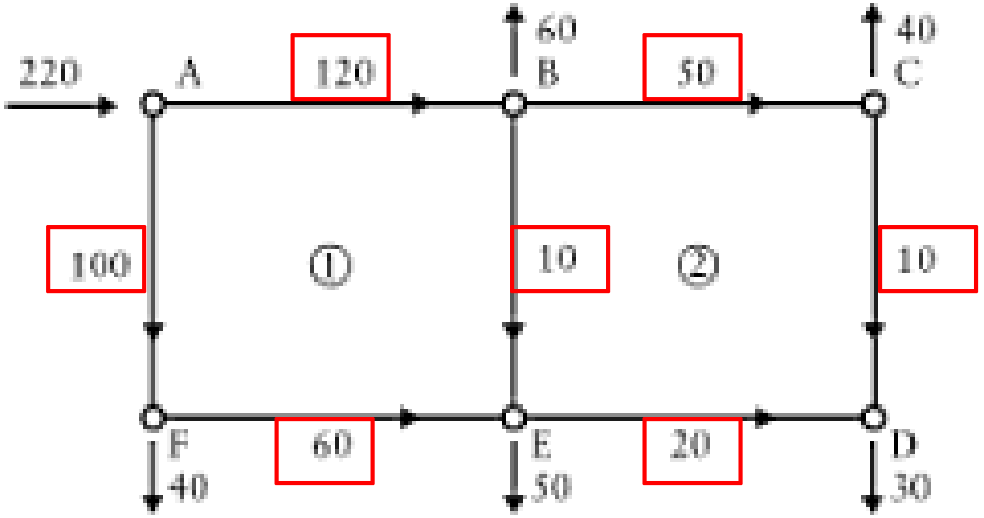
Node	A	B	C	D	E	F
Elevation (m o.d.)	30	25	20	20	22	25


220
A
60
B
40
C
100
①
②
F
40
E
50
D
30


220
A
120
60
B
50
40
C
100
①
10
②
10
F
40
60
E
50
20
D
30

Example: 5-1

- **Solution:**



- Identify loops
  - Loop 1 and Loop 2
- Allocate estimated flows in each pipe
- Compute head loss coefficient of each pipe

$$K = \frac{8fL}{g\pi^2 D^5} \Rightarrow f$$

Moody's Diagram or Colebrook Eq.

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{e/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right)$$

- Compute head loss in each pipe

$$h = h_l = KQ^2$$

Discharge Sign Convention

CW = +ve

CCW = -ve

1st Iteration

Assumed Discharged

	Pipe	e/D	Q (L/s)	Re (x 10^5)	f	K	h_L (m)	h_L/Q $\left(\frac{m}{m^3/s}\right)$
Loop 1	AB	0.00024	120.00	5.41	0.0157	797.0	11.48	95.64
	BE	0.00060	10.00	1.31	0.0205	33 877.0	3.39	338.77
	EF	0.00040	−60.00	4.51	0.0172	11 229.1	−40.42	673.75
	FA	0.00030	−100.00	5.63	0.0162	336.6	−8.36	83.66
						$\sum$	−33.91	1191.82

Correction

$$\Rightarrow \Delta Q = \frac{-\sum h}{2\sum h/Q} = \frac{-(-33.91)}{2 \times 1191.82} = 0.01423 = 14.23 \text{ L/s.}$$

Corrected Discharge

$Q = Q + \Delta Q$

	Pipe	e/D	Q (L/s)
Loop 1	AB	0.00024	120+14.23=134.23
	BE	0.00060	10+14.23=24.23
	EF	0.00040	-60+14.23=-45.77
	FA	0.00030	-100+14.23=-85.77

1st Iteration

Assumed/corrected Discharged

	Pipe	Q (L/s)	Re (x 10^5)	f	K	h_L m)	h_L/Q $\left(\frac{m}{m^3/s}\right)$
Loop 2	BC	50.0	3.76	0.0174	11 359.7	28.40	567.98
	CD	10.0	1.13	0.0205	33 877.0	3.39	338.77
	DE	−20.0	1.50	0.0189	12 338.9	−4.94	246.78
	EB	−24.23	2.73	0.0189	31 232.9	−18.34	756.77
					Σ	8.51	1910.30

Correction

$\Rightarrow \Delta Q = -2.23$ L/s.

Corrected Discharge

$Q = Q + \Delta Q$

	Pipe	Q (L/s)
Loop 2	BC	50-2.23=47.77
	CD	10-2.23=7.77
	DE	-20-2.23=-23.23
	EB	-24.23-2.23=-26.46

2nd Iteration

	Pipe	Q (L/s)	Re (x 10^5)	f	K	h_L (m)	h_L/Q $\left(\frac{m}{m^3/s}\right)$
Loop 1	AB	134.23	6.05	0.0156	791.9	14.27	106.30
	BE	26.46	2.98	0.0188	31 067.7	21.75	822.05
	EF	−45.77	3.44	0.0175	11 424.9	−23.93	522.92
	FA	−85.77	4.83	0.0164	846.9	−6.23	72.64
					$\sum$	5.86	1523.91

$\Rightarrow \Delta Q = -1.92$ L/s.

	Pipe	Q (L/s)
Loop 1	AB	134.23-1.92= 132.31
	BE	26.46-1.92= 24.54
	EF	-45.77-1.92= -47.69
	FA	-85.77-1.92= -87.69

2nd Iteration

- Proceed to loop 2 again and continue iterating until $\sum h_l = 0$

Final values

Pipe	Q (L/s)	h (m)
AB	131.55	13.70
BE	25.02	19.55
FE	48.45	26.67
AF	88.45	6.59
BC	46.53	24.74
CD	6.55	1.52
ED	23.47	6.69

Note: Flows in direction of pipe identifier, e.g. A → B.

Pressure heads

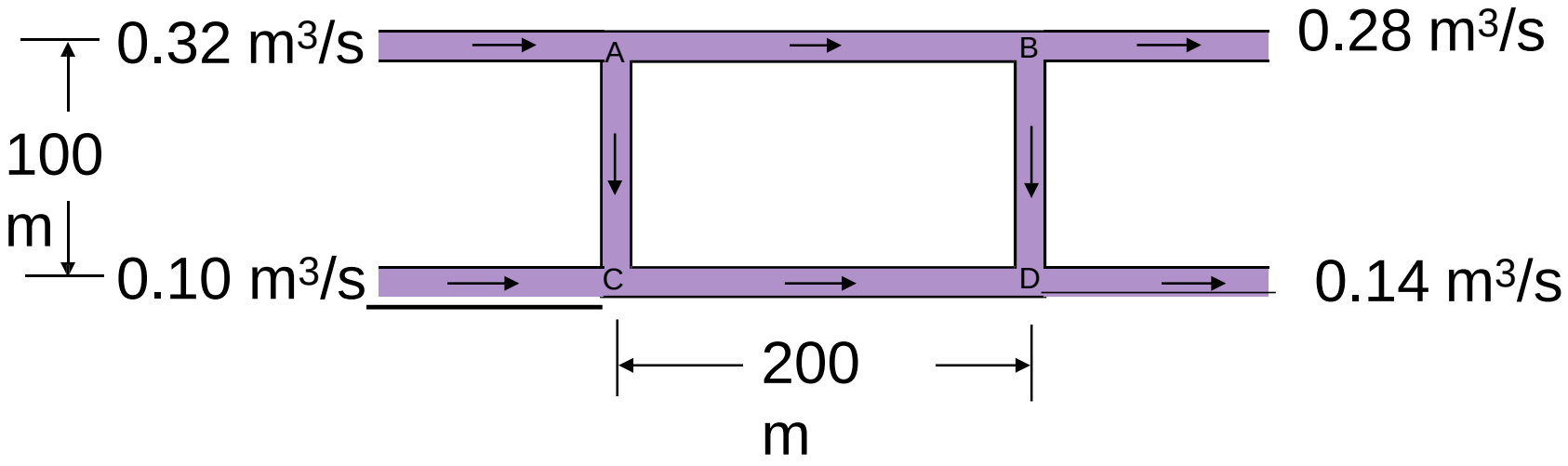
Node	Pressure head (m)
A	40.00
B	31.29
C	11.57
D	10.05
E	14.74
F	38.41

70-30

70-25-6.59

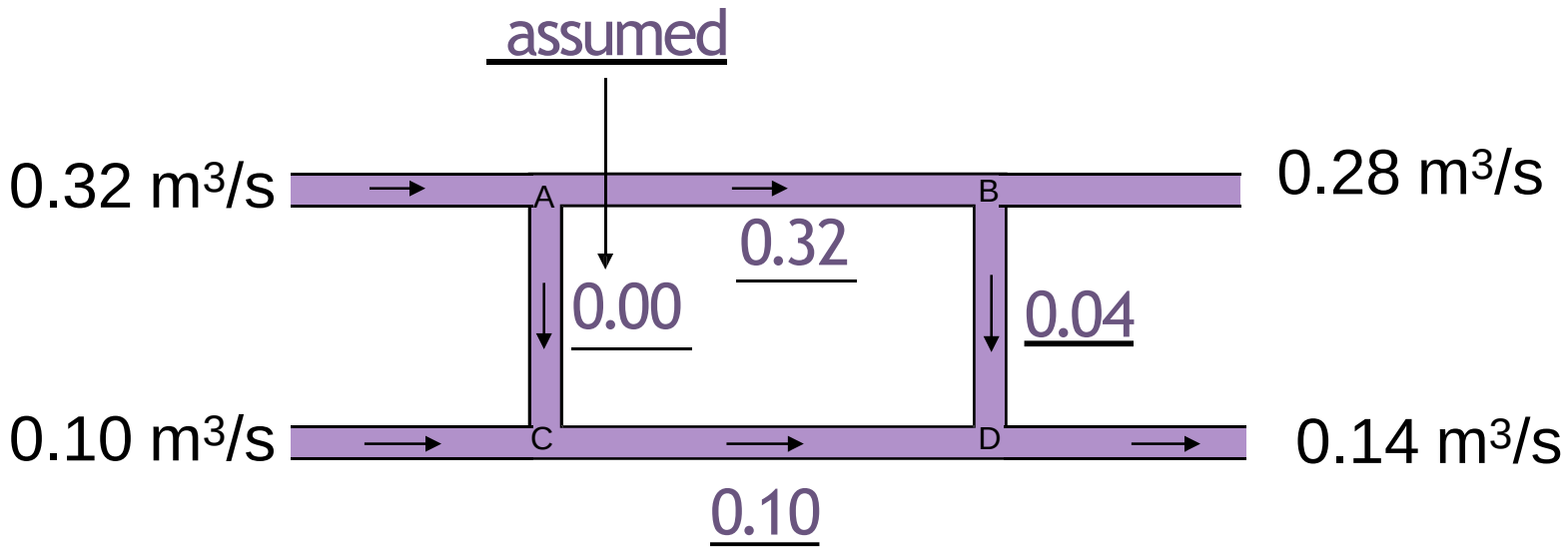
Example:

Find the flows in the loop given the inflows and outflows.
The pipes are all 25 cm cast iron (e=0.26 mm).



Example:

- Identify Loop
- Assign a flow to each pipe link
- Flow into each junction must equal flow out of the junction



Example:

- Calculate the head loss in each pipe

$$h_f = KQ^2$$

$$K = \left| \frac{8fL}{gD^5\pi^2} \right|$$

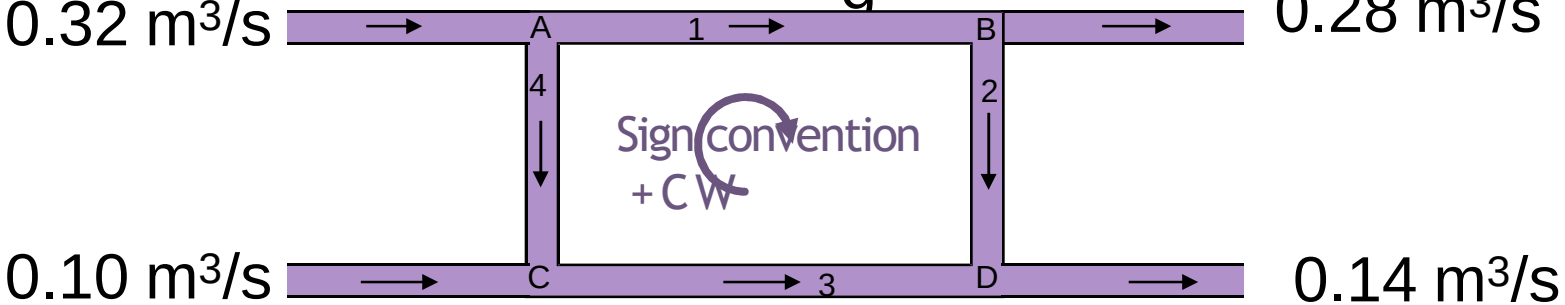
f=0.02 for Re>200000

$$h_{f_1} = 34.7m$$
$$h_{f_2} = 0.222m$$
$$h_{f_3} = -3.39m$$
$$h_{f_4} = -0.00m$$
$$\sum_{i=1}^{4} h_{f_i} = 31.53m$$

$$k_1 = \left| \frac{8(0.02)(200)}{(9.8)(0.25)^5\pi^2} \right| = 339 \frac{s^2}{m^5}$$

$k_1, k_3 = 339$

$k_2, k_4 = 169$



Example:

- The head loss around the loop isn't zero
- Need to change the flow around the loop
  - the clockwise flow is too great (head loss is positive)
  - reduce the clockwise flow to reduce the head loss
- Solution techniques
  - Hardy Cross loop-balancing (optimizes correction)
  - Use a numeric solver (Solver in Excel) to find a change in flow that will give zero head loss around the loop
  - Use Network Analysis software (EPANET, WaterCad, etc.)

Example:

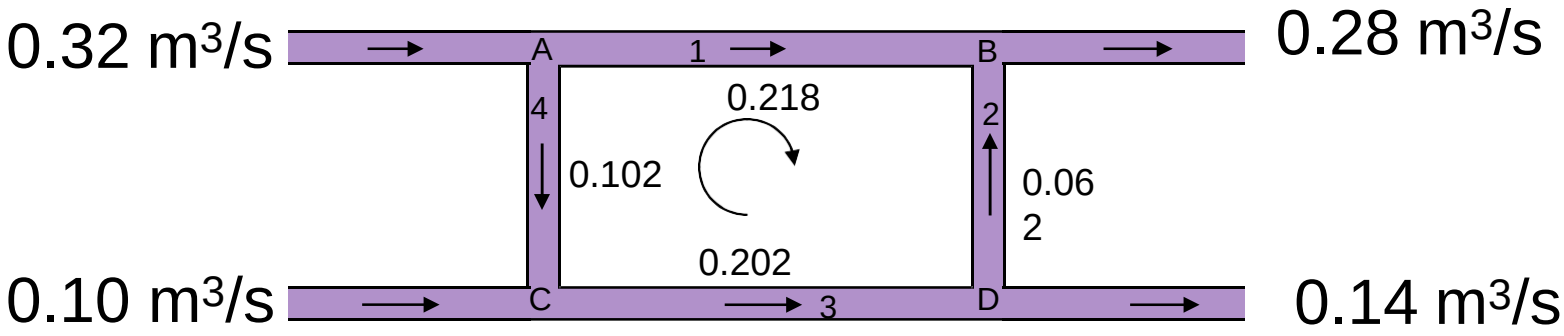
- Numerical Solver
- Set up a spreadsheet as shown below.
- the numbers in bold were entered, the other cells are calculations
- initially ΔQ is 0
- use "solver" to set the sum of the head loss to 0 by changing ΔQ
- the column $Q_0 + \Delta Q$ contains the correct flows

ΔQ	0.000						
pipe	f	L	D	k	Q0	Q0+ΔQ	hf
P1	**0.02**	**200**	**0.25**	339	**0.32**	0.320	34.69
P2	**0.02**	**100**	**0.25**	169	**0.04**	0.040	0.27
P3	**0.02**	**200**	**0.25**	339	**-0.1**	-0.100	-3.39
P4	**0.02**	**100**	**0.25**	169	**0**	0.000	0.00
					Sum Head Loss		31.575

Example

Solution to Loop Problem

$Q_0 + \Delta Q$
0.218
–0.062
–0.202
–0.102

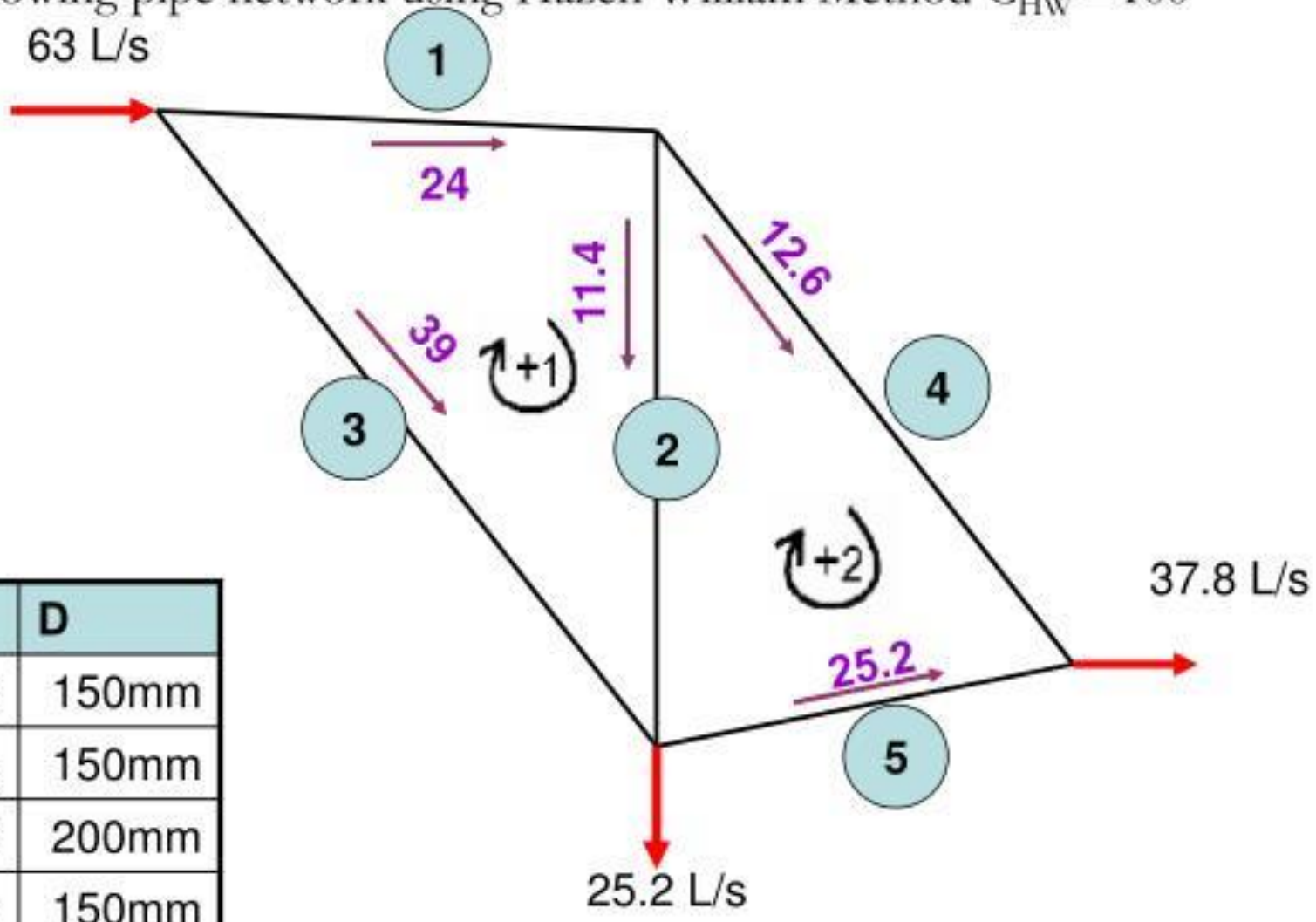


You must be able to make a hand calculation before opting for a better solution using software with a GUI.

Pipe Network Analysis

Example 1

Solve the following pipe network using Hazen William Method C_{HW} =100



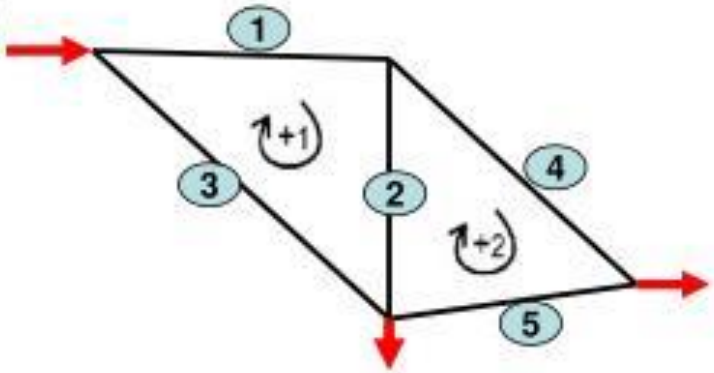
pipe	L	D
1	305m	150mm
2	305m	150mm
3	610m	200mm
4	457m	150mm
5	153m	200mm

$$h_f = \frac{10.7\,L}{C_{HW}^{1.852}\,D^{4.87}}\,Q^{1.852} \Rightarrow C_{HW} = 100, Q = in \text{ L/s}$$

$$h_f = \frac{10.7\,L}{C_{HW}^{1.852}\,D^{4.87}}\left(\frac{Q}{1000}\right)^{1.852}$$

$$h_f = \left\{6.02\times10^{-9}\,\frac{L}{D^{4.87}}\right\}Q^{1.852}$$

$$h_f = \{K\}\,Q^{1.852}$$



1st Iteration

Loop	Pipe	Dia (m)	L (m)	K	Q_o (L/s)	h_f (m)	h_f/Q_o (m/L/s)	Correction L/s	Q L/s
1	1	0.150	305	0.0187	+24.0	+6.68	0.28	-0.24	+23.76
	2	0.150	305	0.0187	+11.4	+1.69	0.15	-0.24+0.57	+11.73
	3	0.200	610	0.0092	-39.0	-8.09	0.21	-0.24	-39.24
						+0.28	0.64		
2	2	0.150	305	0.0187	-11.4	-1.69	0.15	-0.57+0.24	-11.73
	4	0.150	457	0.0280	+12.6	+3.04	0.24	-0.57	+12.03
	5	0.200	153	0.0023	-25.2	-0.90	0.04	-0.57	-25.77
						+0.45	0.43		

$$\Delta_1 = \frac{-\sum h_F}{n\sum \frac{h_F}{Q_o}} = \frac{-0.28}{1.85(0.64)} = -0.24$$

$$\Delta_2 = \frac{-\sum h_F}{n\sum \frac{h_F}{Q_o}} = \frac{-0.45}{1.85(0.43)} = -0.57$$

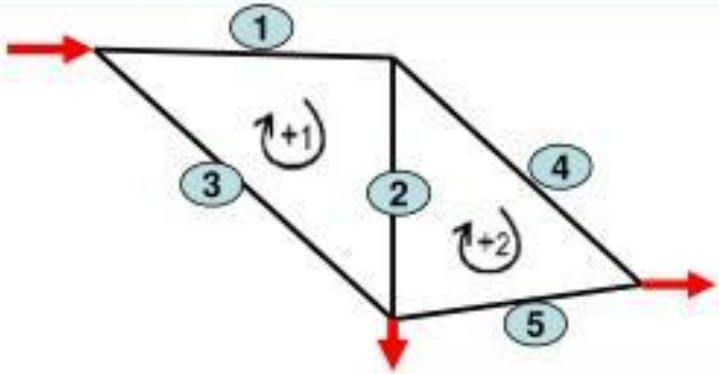
for pipe 2 in loop 1

$\Delta = \Delta_1 - \Delta_2$

for pipe 2 in loop 2

$\Delta = \Delta_2 - \Delta_1$

Pipe Network Analysis



2nd Iteration

Loop	Pipe	Dia (m)	L (m)	K	Q_o (L/s)	h_f (m)	h_f/Q_o (m/L/s)	Correction L/s	Q L/s
1	1	0.150	305	0.0187	+23.76	+6.56	0.28	-0.15	+23.61
	2	0.150	305	0.0187	+11.73	+1.79	0.15	-0.15+0.09	+11.67
	3	0.200	610	0.0092	-39.24	-8.17	0.21	-0.15	-39.39
						+0.18	0.64		
2	2	0.150	305	0.0187	-11.73	-1.78	0.15	-0.09+0.15	-11.67
	4	0.150	457	0.0280	+12.03	+2.79	0.23	-0.09	+11.94
	5	0.200	153	0.0023	-25.77	-0.94	0.04	-0.09	-25.86
						+0.07	0.42		

$$\Delta_1 = \frac{-\sum h_F}{n\sum \frac{h_F}{Q_o}} = \frac{-0.18}{1.85(0.64)} = -0.15$$

$$\Delta_1 = \frac{-\sum h_F}{n\sum \frac{h_F}{Q_o}} = \frac{-(0.07)}{1.85(0.42)} = -0.09$$

for pipe 2 in loop 1

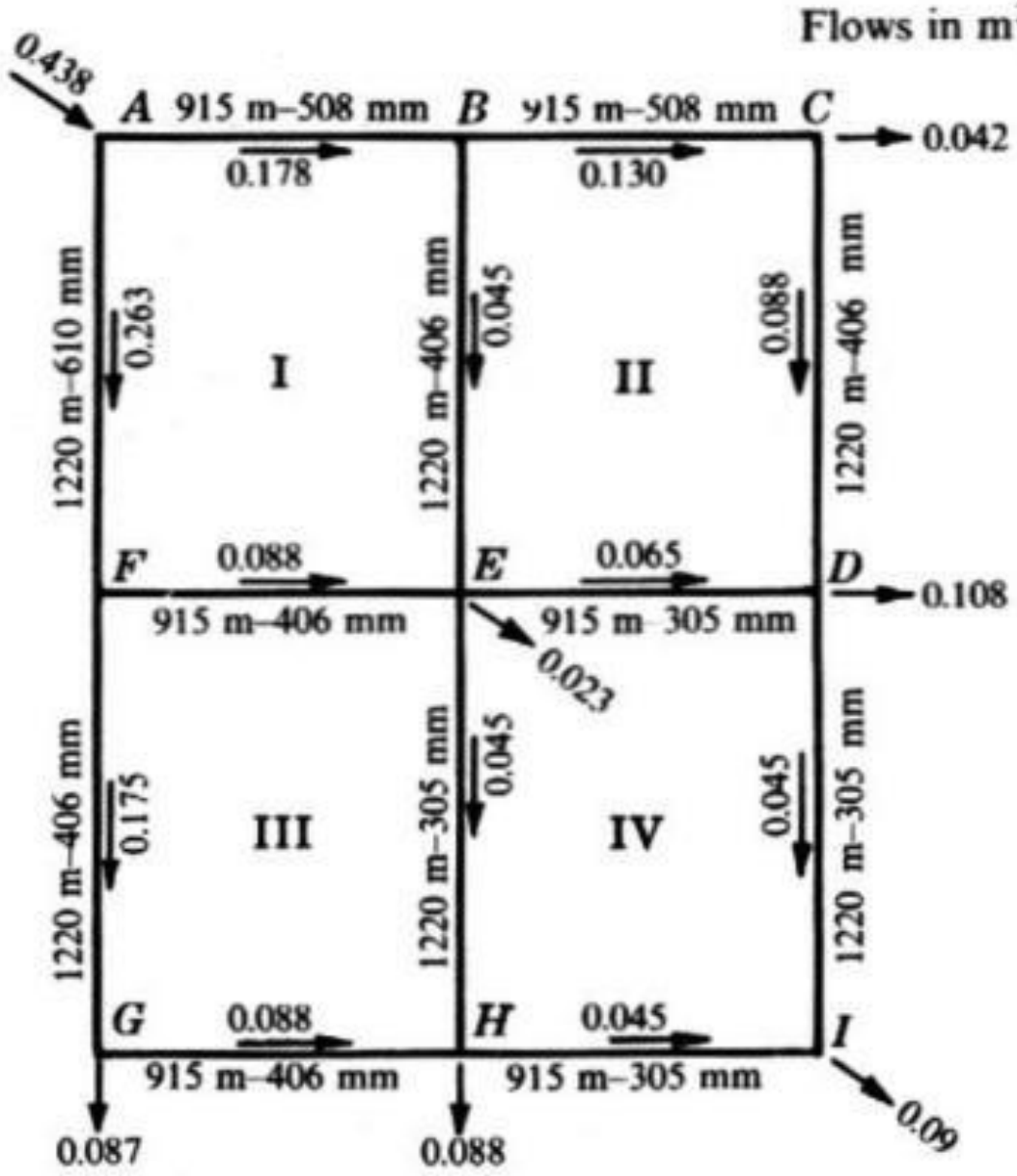
$$\Delta = \Delta_1 - \Delta_2$$

for pipe 2 in loop 2

$$\Delta = \Delta_2 - \Delta_1$$

Pipe Network Analysis

Example 2



Solve the following pipe network using Hazen William Method C_{HW} =120

Iteration 1

Pipe	D(mm)	L(m)	Q_1 (m^3/s)	i m/1000 m	hf, m	$\frac{hf}{Q_1}$	Δ	Q_2
AB	508	915	0.175	1.62	1.48	8.46	+0.014	0.189
BE	406	1220	0.045	0.37	0.45	10.00	+0.014 − 0.006 = +0.008	0.053
EF	406	915	−0.088	−1.33	−1.22	13.86	+0.014 − 0.022 = −0.008	−0.096
FA	610	1220	−0.263	−1.41	−1.72	6.54	+0.014	−0.249
					Σ = −1.01	38.86		
BC	508	915	0.130	0.95	0.87	6.69	+0.006	0.136
CD	406	1220	0.088	1.33	1.62	18.41	+0.006	0.094
DE	305	915	−0.065	−3.15	−2.88	44.31	+0.006 − (−0.005) = +0.011	−0.054
EB	406	1220	−0.045	−0.37	−0.45	10.00	+0.006 − (0.014) = −0.008	−0.053
					Σ = −0.84	79.41		
FE	406	915	0.088	1.33	1.22	13.86	+0.022 − (0.014) = +0.008	0.096
EH	305	1220	0.045	1.48	1.81	40.22	+0.022 − (−0.005) = +0.027	0.072
HG	406	915	−0.088	−1.33	−1.22	13.86	+0.022	−0.066
GF	406	1220	−0.175	−4.85	−5.91	33.77	+0.022	−0.153
					Σ = −4.1	101.71		
ED	305	915	0.065	3.15	2.88	44.31	−0.005 − (0.006) = −0.011	0.054
DI	305	1220	0.045	1.48	1.81	40.22	−0.005	0.040
IH	305	915	−0.045	−1.48	−1.35	30.00	−0.005	−0.050
HE	305	1220	−0.045	−1.48	−1.81	40.22	−0.005 − (0.022) = −0.027	−0.072
					Σ = +1.53	154.75		

Pipe Network Analysis

for Iteration 1

$$\Delta = \frac{-\sum kQ_o^n}{\sum nkQ_o^{(n-1)}} = \frac{-\sum h_f}{n\sum \frac{h_f}{Q_o}}$$

$$\Delta_1 = \frac{-(-1.01)}{1.85(38.86)} = +0.014$$

$$\Delta_2 = \frac{-(-0.84)}{1.85(79.41)} = +0.006$$

$$\Delta_3 = \frac{-(-4.1)}{1.85(101.71)} = +0.022$$

$$\Delta_4 = \frac{-(+1.53)}{1.85(154.75)} = -0.005$$

Iteration 2

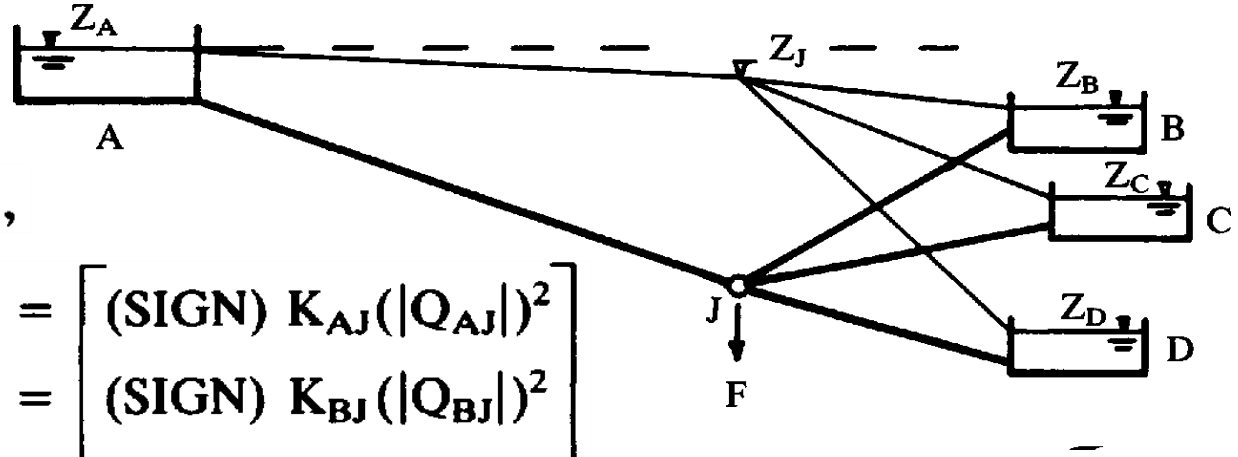
الأنبوب	Q_2	i	hf	hf/Q	Δ
AB	0.189	1.86	1.70	8.99	+0.009
BE	0.053	0.51	0.62	11.70	+0.009 + negl = +0.009
EF	−0.096	−1.57	−1.44	15.00	+0.009 − (−0.003) = +0.012
FA	−0.249	−1.28	−1.56	6.27	+0.009
			Σ = −0.68	41.96	
BC	0.136	1.02	0.93	6.84	negl
CD	0.094	1.48	1.80	19.15	negl
DE	−0.054	−2.28	−2.08	38.52	negl − 0.008 = −0.008
EB	−0.053	−0.51	−0.62	11.70	negl − 0.009 = −0.009
			Σ = +0.03	76.21	
FE	0.096	1.57	1.44	15.00	−0.003 − 0.009 = −0.012
EH	0.072	3.65	4.45	61.80	−0.003 − 0.008 = −0.011
HG	−0.066	−0.79	−0.72	10.91	−0.003
GF	−0.153	−3.75	−4.57	29.87	−0.003
			Σ = +0.6	117.58	
ED	0.054	2.28	2.08	38.52	+0.008 + negl = +0.008
DI	0.040	1.18	1.44	36.00	+0.008
IH	−0.050	−1.83	−1.67	33.40	+0.008
HE	−0.072	−3.65	−4.45	61.81	+0.008 − (−0.003) = +0.011
			Σ = −2.60	169.73	

Iteration 3

الأنبوب	Q_3	i	hf	hf/Q	Δ	Q_4
AB	0.198	2.02	1.85	9.34	−0.001	0.197
BE	0.062	0.68	0.83	13.39	−0.001 − 0.005 = −0.006	0.056
EF	−0.084	−1.25	−1.14	13.57	−0.001 − 0.005 = −0.006	−0.090
FA	−0.240	−1.20	−1.46	6.08	−0.001	−0.241
			Σ = +0.08	42.38		
BC	0.136	1.02	0.93	6.84	+0.005	0.141
CD	0.094	1.49	1.82	19.36	+0.005	0.099
DE	−0.062	−2.97	−2.72	43.87	+0.005 + 0.001 = +0.006	−0.056
EB	−0.062	−0.68	−0.83	13.39	+0.005 + 0.001 = +0.006	−0.056
			Σ = −0.8	83.46		
FE	0.084	1.25	1.14	13.57	+0.005 + 0.001 = +0.006	0.090
EH	0.061	2.68	3.27	53.61	+0.005 + 0.001 = +0.006	0.067
HG	−0.069	−0.84	−0.77	11.16	+0.005	−0.064
GF	−0.156	−3.90	−4.75	30.45	+0.005	−0.151
			Σ = −1.11	108.79		
ED	0.062	2.97	2.72	43.87	−0.001 − 0.005 = −0.006	0.056
DI	0.048	1.68	2.05	42.71	−0.001	0.047
IH	−0.042	−1.31	−1.20	28.57	−0.001	−0.043
HE	−0.061	−2.68	−3.27	53.61	−0.001 − 0.005 = −0.006	−0.067
			Σ = +0.30	168.76		

(2). Nodal Method

- Fig shows a branched pipe system delivering water from impounding reservoir A to the service reservoirs B, C and D. F is known direct out flow from J.



$h_L = KQ^2$,

$$\begin{bmatrix} Z_A - Z_J \\ Z_B - Z_J \\ \cdot \quad \cdot \\ \cdot \quad \cdot \\ Z_I - Z_J \end{bmatrix} = \begin{bmatrix} (\text{SIGN})\ K_{AJ}(|Q_{AJ}|)^2 \\ (\text{SIGN})\ K_{BJ}(|Q_{BJ}|)^2 \\ \cdot \quad \cdot \quad \cdot \\ \cdot \quad \cdot \quad \cdot \\ (\text{SIGN})\ K_{IJ}(|Q_{IJ}|)^2 \end{bmatrix} \qquad \text{Eq. (1)}$$

(SIGN) is + or − according to the sign of $(Z_I - Z_J)$. Thus flows towards the junction are positive and flows away from the junction are negative.

(2). Nodal Method

The continuity equation for flow rates at J is:

$$\Sigma\, Q_{IJ} - F\ (= Q_{AJ} + Q_{BJ} + Q_{CJ} + Q_{DJ} - F) = 0 \qquad \text{Eq. (2)}$$

- Eq. (1) can be written as

$$[Q_{IJ}] = \left[(\text{SIGN}) \left(\frac{|Z_I - Z_J|}{K_{IJ}}\right)^{1/2}\right] \qquad \text{Eq. (3)}$$

Iteration of Z_J can be performed such that Q_{IJ} from Eq. (3) satisfies Eq. (2).

If $(\sum Q_{IJ}-F)\neq 0$ then a correction, ΔZ_J, is applied to Z_J such that

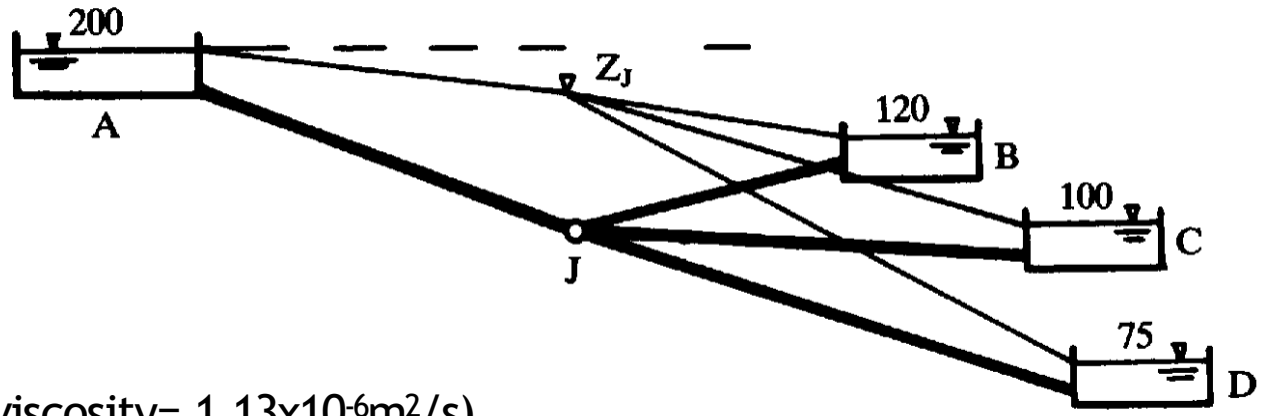
$$\Delta Z_J = \frac{2(\Sigma Q_{IJ} - F_J)}{\sum \frac{Q_{IJ}}{h_{L,IJ}}}$$

Example 5-4

Determine the discharges in the pipes of the network shown in fig. neglecting minor losses.

Pipe	Length (m)	Diameter (mm)
AJ	10 000	450
BJ	2000	350
CJ	3000	300
DJ	3000	250

Roughness size of all pipes = 0·06 mm



(kinematic viscosity= $1.13 \times 10^{-6} m^2/s$)

The friction factor f may be obtained from the Moody diagram.

Example 5-4

Solution:

Estimate Z_J (pressure head elevation at J) = 150·0 m a.o.d.

1st correction

Pipe No. (I–J)	Velocity (estimate) m/s	Re($\times 10^5$)	f	K	$Z_I - Z_J$	Q (m^3/s)	$(Q/h)_L \times 10^{-3}$	Q/A (m/s)
AJ	2·0	7·96	0·0145	649	+50	0·2775	5·55	1·75
BJ	2·0	6·20	0·0150	472	−30	−0·2521	8·40	2·62
CJ	2·0	5·31	0·0155	1581	−50	−0·1778	3·56	2·50
DJ	2·0	4·42	0·0165	4188	−75	−0·1338	1·78	2·73
					Σ	−0·2862	0·0193	

Correction to $Z_J = \dfrac{2(-0{\cdot}2862)}{0{\cdot}0193} = -29{\cdot}67$: $Z_J = 120{\cdot}33$ m

Example 5-4

Final values:

$Q_{AJ} = 0{\cdot}344\ m^3/s$; $Q_{JB} = 0{\cdot}105\ m^3/s$; $Q_{JC} = 0{\cdot}127\ m^3/s$;

$Q_{JD} = 0{\cdot}112\ m^3/s$.

Newton Raphson Method

- Newton-Raphson method is a powerful numerical method for solving systems of nonlinear equations.

- Suppose that there are three nonlinear equations $F_1(Q_1, Q_2, Q_3) = 0$, $F_2(Q_1, Q_2, Q_3) = 0$, and $F_3(Q_1, Q_2, Q_3) = 0$ to be solved for Q_1, Q_2, and Q_3.

- Adopt a starting solution (Q_1, Q_2, Q_3).

- Also consider that $(Q_1 + \Delta Q_1, Q_2, + \Delta Q_2, Q_3 + \Delta Q_3)$ is the solution of the set of equations. That is

$$F_1(Q_1 + \Delta Q_1, Q_2 + \Delta Q_2, Q_3 + \Delta Q_3) = 0$$
$$F_2(Q_1 + \Delta Q_1, Q_2 + \Delta Q_2, Q_3 + \Delta Q_3) = 0$$
$$F_3(Q_1 + \Delta Q_1, Q_2 + \Delta Q_2, Q_3 + \Delta Q_3) = 0. \qquad \text{Eq. (1)}$$

Newton Raphson Method

Expanding the above equations as Taylor's series,

$$
\begin{aligned}
F_1 + [\partial F_1/\partial Q_1]\Delta Q_1 + [\partial F_1/\partial Q_2]\Delta Q_2 + [\partial F_1/\partial Q_3]\Delta Q_3 = 0 \\
F_2 + [\partial F_2/\partial Q_1]\Delta Q_1 + [\partial F_2/\partial Q_2]\Delta Q_2 + [\partial F_2/\partial Q_3]\Delta Q_3 = 0 \\
F_3 + [\partial F_3/\partial Q_1]\Delta Q_1 + [\partial F_3/\partial Q_2]\Delta Q_2 + [\partial F_3/\partial Q_3]\Delta Q_3 = 0.
\end{aligned}
\quad \text{Eq. (2)}
$$

Arranging the above set of equations in matrix form,

$$
\begin{bmatrix}
\partial F_1/\partial Q_1 & \partial F_1/\partial Q_2 & \partial F_1/\partial Q_3 \\
\partial F_2/\partial Q_1 & \partial F_2/\partial Q_2 & \partial F_2/\partial Q_3 \\
\partial F_3/\partial Q_1 & \partial F_3/\partial Q_2 & \partial F_3/\partial Q_3
\end{bmatrix}
\begin{bmatrix}
\Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3
\end{bmatrix}
= -
\begin{bmatrix}
F_1 \\ F_2 \\ F_3
\end{bmatrix}.
\quad \text{Eq. (3)}
$$

Solving Eq (3)., we get

$$
\begin{bmatrix}
\Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3
\end{bmatrix}
= -
\begin{bmatrix}
\partial F_1/\partial Q_1 & \partial F_1/\partial Q_2 & \partial F_1/\partial Q_3 \\
\partial F_2/\partial Q_1 & \partial F_2/\partial Q_2 & \partial F_2/\partial Q_3 \\
\partial F_3/\partial Q_1 & \partial F_3/\partial Q_2 & \partial F_3/\partial Q_3
\end{bmatrix}^{-1}
\begin{bmatrix}
F_1 \\ F_2 \\ F_3
\end{bmatrix}
\quad \text{Eq. (4)}
$$

Newton Raphson Method

Knowing the corrections, the discharges are improved as

$$\begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} = \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} + \begin{bmatrix} \Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3 \end{bmatrix} \qquad \text{Eq. (5)}$$

Procedure

- The overall procedure for looped network analysis by the Newton-Raphson method can be summarized in the following steps:

- Step 1: Number all the nodes, pipe links, and loops.
- Step 2:Write nodal discharge equations as

$$F_j = \sum_{n=1}^{j_n} Q_{jn} - q_j = 0 \quad \text{for all nodes} - 1,$$

- where Q_{jn} is the discharge in n^{th} pipe at node j, q_j is nodal withdrawal, and jn is the total number of pipes at node j.
- Step 3:Write loop head-loss equations as

$$F_k = \sum_{n=1}^{k_n} K_n Q_{kn} |Q_{kn}| = 0 \quad \text{for all the loops } (n = 1, k_n).$$

- where k_n is total pipes in k_{th} loop.

Procedure

- Step 4: Assume initial pipe discharges Q_1, Q_2, and Q_3., . . . satisfying continuity equations.

Step 5: Determine friction factors, f_i, in all pipe links and compute corresponding K_i using

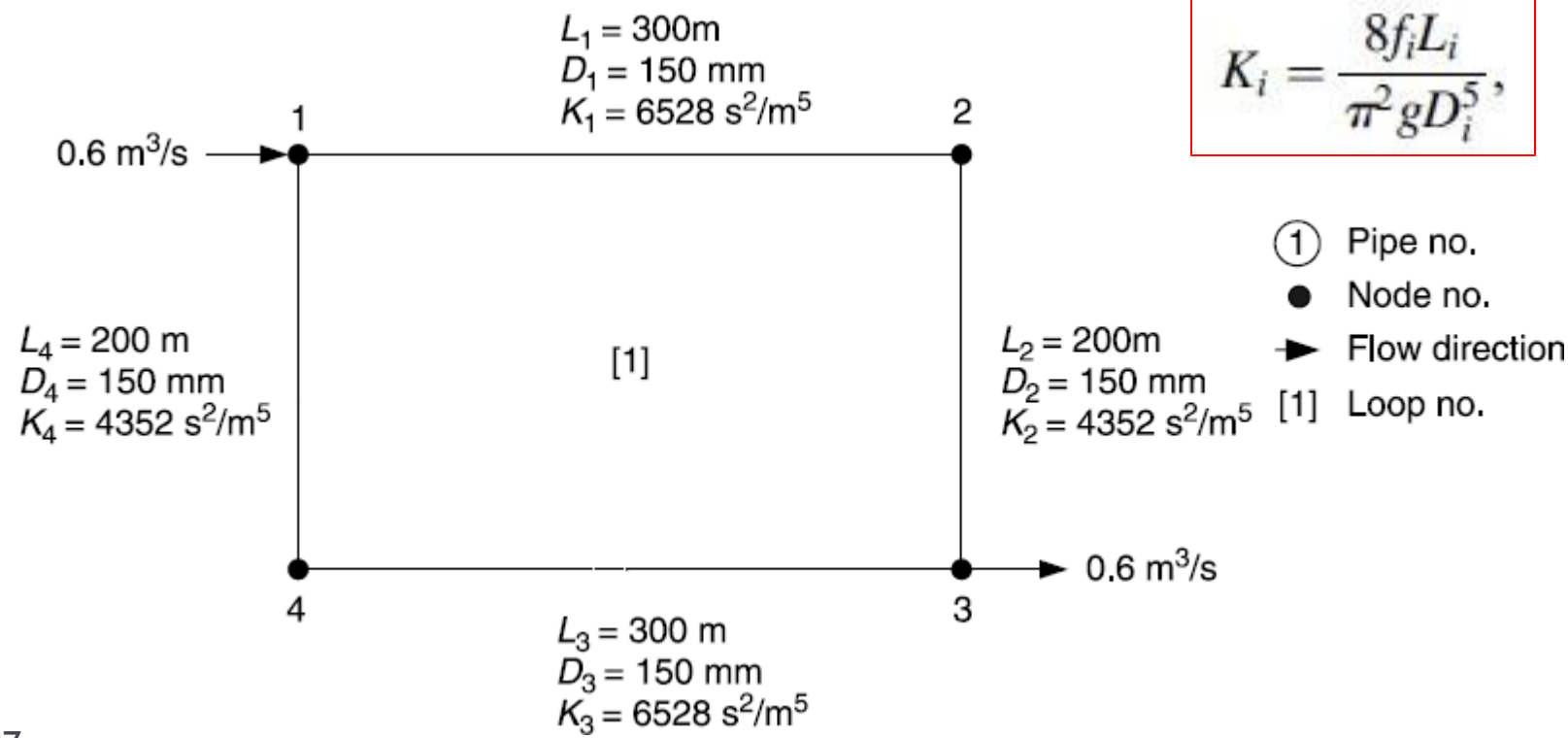
$$K_i = \frac{8 f_i L_i}{\pi^2 g D_i^5},$$

- Step 6: Find values of partial derivatives $\partial F_n / \partial Q_i$ and functions F_n, using the initial pipe discharges Q_i and K_i.
- Step 7: Find ΔQi. The equations generated are of the form $Ax = b$, which can be solved for ΔQ_i.
- Step 8: Using the obtained ΔQ_i values, the pipe discharges are modified and the process is repeated again until the calculated ΔQ_i values are very small.

Example

- The pipe network of two loops as shown in Fig. has to be analyzed by the Newton Raphson method for pipe flows for given pipe lengths L and pipe diameters D.
- The nodal inflow at node 1 and nodal outflow at node 3 are shown in the figure. Assume a constant friction factor f = 0.02.

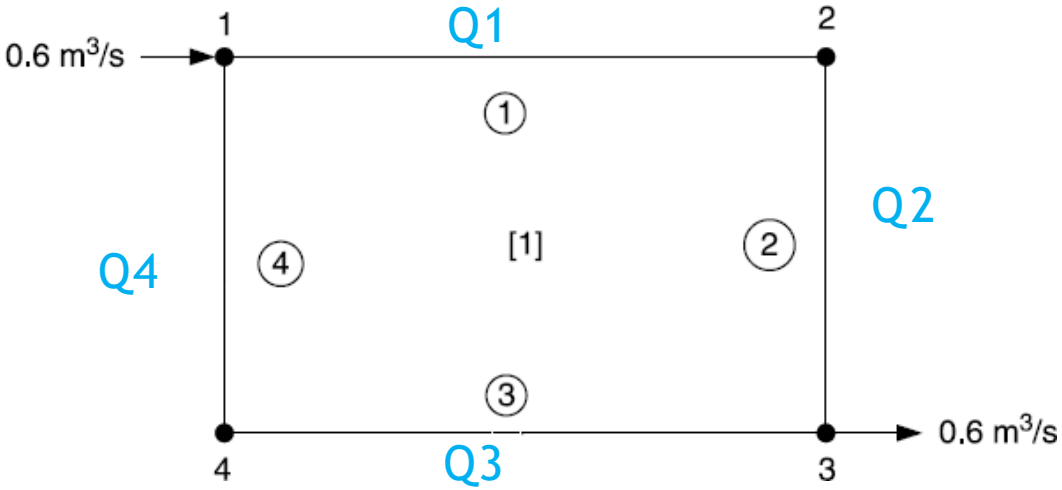
$$K_i = \frac{8 f_i L_i}{\pi^2 g D_i^5},$$



Single looped network

Example

- **Solution**: Write nodal and loop equations



- Node1: $Q_1+Q_4-0.6=0$
- Node 2: $Q_1+Q_2=0$
- Node 3: $Q_2+Q_3-0.6=0$
- Node 4: !!!
- and loop Eq.

$$K_1Q_{1_2}+\ K_2Q_{2_2}+\ K_3Q_{3_2}+\ K_4Q_{4_2}=0$$

$$h_L\,(pipe) = KQ^n$$

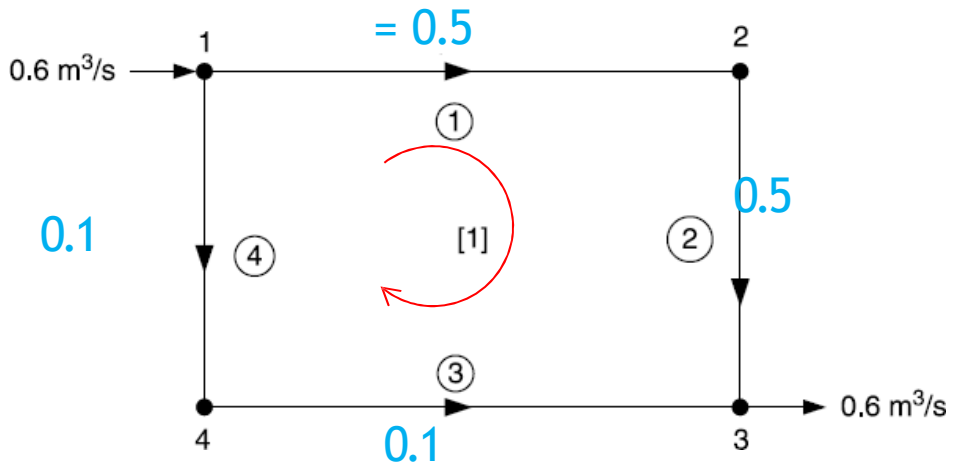
$$n = 2 \quad K\,\frac{8fL}{g\pi^2D^5}$$

Example

- Assume initial pipe discharges based on continuity euqtion:

  $Q_1 = 0.5\ m^3/s$

$$Q_2 = 0.5\ \mathrm{m}^3/\mathrm{s}$$
$$Q_3 = 0.1\ \mathrm{m}^3/\mathrm{s}$$
$$Q_4 = 0.1\ \mathrm{m}^3/\mathrm{s}$$



- Apply signs to flowrate in eqns
  - Nodal discharge functions, F,are
- $Q_1+Q_4-0.6=F_1$
- $-Q_1+Q_2=F_2$
- $Q_2+Q_3-0.6=F_3$
  - Loop head-loss function
- $K_1Q_1^2+K_2Q_2^2-K_3Q_3^2-K_4Q_4^2=F_4$

Apply signs (+ or -)

Example

- The final nodal discharge functions, F, become

$$F_1 = Q_1 + Q_4 - 0.6$$
$$F_2 = -Q_1 + Q_2$$
$$F_3 = Q_2 + Q_3 - 0.6$$

- and loop head-loss function is
- $F_4 = 6528Q_1^2 + 4352Q_2^2 - 6528Q_3^2 - 4352Q_4^2 = 0$

- Take derivatives of equations:

$$\partial F_1/\partial Q_1 = 1 \quad \partial F_1/\partial Q_2 = 0 \quad \partial F_1/\partial Q_3 = 0 \quad \partial F_1/\partial Q_4 = 1$$
$$\partial F_2/\partial Q_1 = -1 \quad \partial F_2/\partial Q_2 = 1 \quad \partial F_2/\partial Q_3 = 0 \quad \partial F_2/\partial Q_4 = 0$$
$$\partial F_3/\partial Q_1 = 0 \quad \partial F_3/\partial Q_2 = 1 \quad \partial F_3/\partial Q_3 = 1 \quad \partial F_3/\partial Q_4 = 0$$
$$\partial F_4/\partial Q_1 = 13056Q_1 \quad \partial F_4/\partial Q_2 = 8704Q_2 \quad \partial F_4/\partial Q_3 = -13056Q_3 \quad \partial F_4/\partial Q_4 = -8704Q_4$$

Example

- Assembled equation in the following matrix form:

$$\begin{bmatrix} \Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3 \\ \Delta Q_4 \end{bmatrix} = - \begin{bmatrix} \partial F_1/\partial Q_1 & \partial F_1/\partial Q_2 & \partial F_1/\partial Q_3 & \partial F_1/\partial Q_4 \\ \partial F_2/\partial Q_1 & \partial F_2/\partial Q_2 & \partial F_2/\partial Q_3 & \partial F_2/\partial Q_4 \\ \partial F_3/\partial Q_1 & \partial F_3/\partial Q_2 & \partial F_3/\partial Q_3 & \partial F_3/\partial Q_4 \\ \partial F_4/\partial Q_1 & \partial F_4/\partial Q_2 & \partial F_4/\partial Q_3 & \partial F_4/\partial Q_4 \end{bmatrix}^{-1} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

- Substitute the derivatives

$$\begin{bmatrix} \Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3 \\ \Delta Q_4 \end{bmatrix} = - \begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ \mathbf{13056Q_1} & \mathbf{8704Q_2} & \mathbf{-13056Q_3} & \mathbf{-8704Q_4} \end{bmatrix}^{-1} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

Example

- Now, calculate values of F_1 to F_4 using previously assumed flows i.e.,
- $Q_1 = 0.5\ m^3/s$

$$Q_2 = 0.5\ \mathrm{m}^3/\mathrm{s}$$
$$Q_3 = 0.1\ \mathrm{m}^3/\mathrm{s}$$
$$Q_4 = 0.1\ \mathrm{m}^3/\mathrm{s}$$

$$F_1 = Q_1 + Q_4 - 0.6 = 0$$
$$F_2 = -Q_1 + Q_2 = 0$$
$$F_3 = Q_2 + Q_3 - 0.6 = 0$$
$$F_4 - 0.528Q_{1_2} + 4352Q_{2_2} - 0.528Q_{3_2} - 4352Q_{4_2} = 2611.2$$

Example

- Substitute values of F_1 to F_4 and Q_1 to Q_4 in the matrix form :

$$\begin{bmatrix} \Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3 \\ \Delta Q_4 \end{bmatrix} = -\begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ \mathbf{6528} & \mathbf{4352} & \mathbf{-1305.6} & \mathbf{-870.4} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2611.2 \end{bmatrix}$$

- Solve the matrix, we get following flowrate corrections:

$$\Delta Q_1 = -0.2\ \mathrm{m}^3/\mathrm{s}$$
$$\Delta Q_2 = -0.2\ \mathrm{m}^3/\mathrm{s}$$
$$\Delta Q_3 = 0.2\ \mathrm{m}^3/\mathrm{s}$$
$$\Delta Q_4 = 0.2\ \mathrm{m}^3/\mathrm{s}$$

Example

- Using discharge corrections, the revised pipe flowrates become:

$$Q_1 = Q_1 + \Delta Q_1 = 0.5 - 0.2 = 0.3\ \mathrm{m}^3/\mathrm{s}$$
$$Q_2 = Q_2 + \Delta Q_2 = 0.5 - 0.2 = 0.3\ \mathrm{m}^3/\mathrm{s}$$
$$Q_3 = Q_3 + \Delta Q_3 = 0.1 + 0.2 = 0.3\ \mathrm{m}^3/\mathrm{s}$$
$$Q_4 = Q_4 + \Delta Q_4 = 0.1 + 0.2 = 0.3\ \mathrm{m}^3/\mathrm{s}$$

- The process is repeated with the new pipe flowrates. Revised values of F and derivative ∂F/∂Q values are obtained. Substituting the revised values, the following new solution matrix is generated:

$$\begin{bmatrix} \Delta Q_1 \\ \Delta Q_2 \\ \Delta Q_3 \\ \Delta Q_4 \end{bmatrix} = -\begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ \mathbf{3916.8} & \mathbf{2611.2} & \mathbf{-3916.8} & \mathbf{-2611.2} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Example

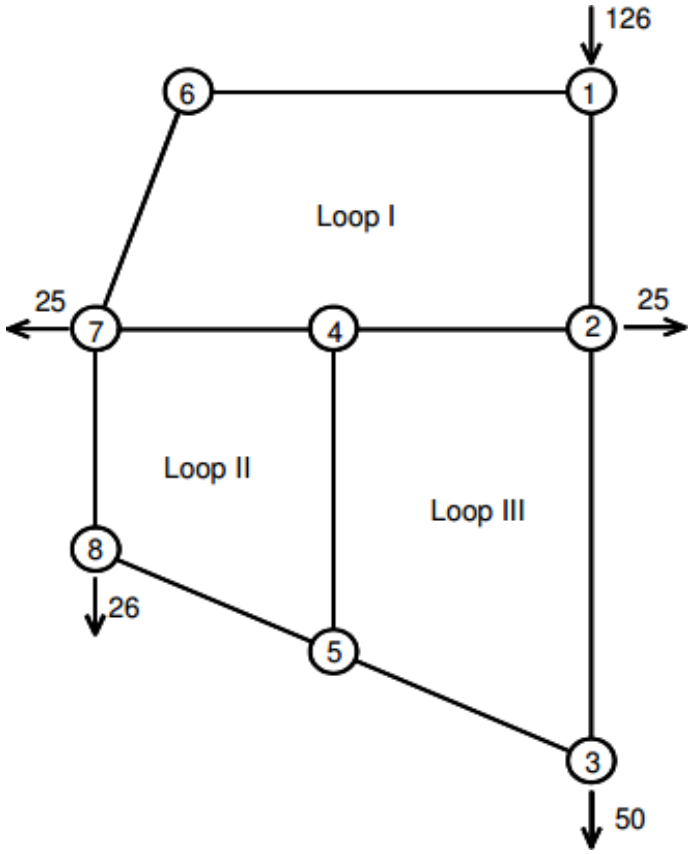
- As the right-hand side is operated upon null vector, all the discharge corrections $\Delta Q = 0$.
- Thus, the final flowrates are

$$Q_1 = 0.3\ \mathrm{m}^3/\mathrm{s}$$
$$Q_2 = 0.3\ \mathrm{m}^3/\mathrm{s}$$
$$Q_3 = 0.3\ \mathrm{m}^3/\mathrm{s}$$
$$Q_4 = 0.3\ \mathrm{m}^3/\mathrm{s}$$

Problem

- Write solution matrix for analysis using Newton Raphson Method

- Node Eqns.=No. ofnodes-1
- Loop Eqns. =3



Problem: Solution

- Nodal and loops equation in their general form are given below

Node equations

Node	Equation
1	$126 + Q_{12} + Q_{16} = 0$
2	$-25 + Q_{21} + Q_{23} + Q_{24} = 0$
3	$-50 + Q_{35} + Q_{32} = 0$
4	$Q_{45} + Q_{47} + Q_{42} = 0$
5	$Q_{58} + Q_{54} + Q_{53} = 0$
6	$Q_{67} + Q_{61} = 0$
7	$-25 + Q_{76} + Q_{74} + Q_{78} = 0$

Discharge_in=+ve

Discharge_out=-ve

Loop equations

Loop	Equation
1	$h_{12} + h_{24} + h_{47} + h_{76} + h_{61} = 0$
2	$h_{23} + h_{35} + h_{54} + h_{42} = 0$
3	$H_{45} + h_{58} + h_{87} + h_{74} = 0$

CW=+ve

CCW=-ve

Thank You